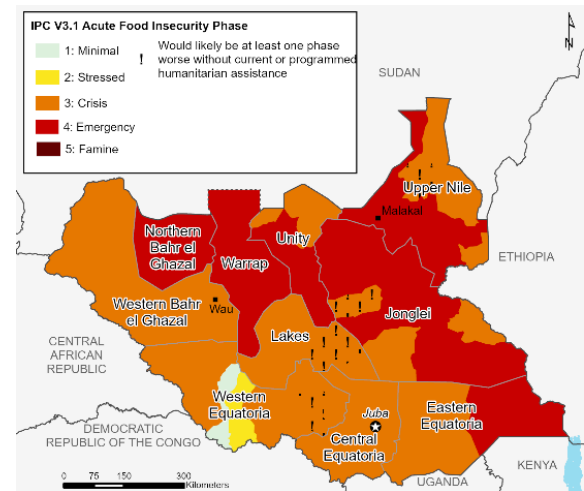


*Protracted conflict and floods drive widespread Emergency (IPC Phase 4) during the harvesting period*

**KEY MESSAGES**

- Extensive flooding for the fourth consecutive year has affected over a million people, caused crop and livestock production losses, significantly disrupted trade flows and market functionality, and impeded the delivery of humanitarian food and nutrition assistance to vulnerable people in need. This shock coupled with ongoing conflict come at a time of deteriorating economic conditions characterized by currency depreciation and high food prices, all of which are driving persistently high levels of acute food insecurity. Crisis (IPC Phase 3) and Emergency (IPC Phase 4) outcomes are expected to remain widespread even during the harvest (October-January) and post-harvest (February-March) periods. While humanitarians plan to deliver food assistance to 1.3 million people per month on average, this represents only 16 percent of the total population in need.
- The largest numbers of flood-affected people are found in Northern Bahr el Ghazal, Warrap, Upper Nile, and Unity. Recent conflict hotspots include northern Jonglei and south-western Upper Nile, with simmering insecurity in central and northern Unity, Greater Tonj and Twic of Warrap, and parts of Greater Equatoria. Of particular concern are populations in areas with concurrent insecurity and floods, notably northern Jonglei and central Unity. FEWS NET and [IPC partners](#) assess that the worst-affected households in Fangak, Canal/Pigi, Akobo, and Pibor, who lack productive assets and face physical or other barriers to accessing food assistance, are in Catastrophe (IPC Phase 5).
- Due to the above shocks, as well as prolonged dry spells in north-central and southern South Sudan and limited household access to inputs, harvests in Greater Upper Nile and Greater Bahr el Ghazal are expected to be lower than last year and the five-year average. By contrast, the second season harvest is likely to be higher than last year's in bi-modal Greater Equatoria due to relative calm in productive cropping zones. Overall, the resultant national cereal deficit and high costs of regional food imports will drive high prices and limit food access.
- While the main dry season (November-May) will facilitate receding floodwaters, flood extent will remain atypically expansive as has been observed over the past few years. In addition, levels of conflict and insecurity typically increase during the dry season. Millions of households will be unable to return to their places of origin or recover their typical livelihoods. As a result, the availability of the harvest will only marginally reduce the severity and magnitude of acute food insecurity. As food stocks are depleted in early 2023, and livestock productivity reaches an annual low between February and May, millions will face food consumption deficits or rely on severe coping strategies to access food.
- FEWS NET assesses that a credible [risk of Famine \(IPC Phase 5\)](#) persists in Jonglei and Unity, where there are large populations in Emergency (IPC Phase 4) with associated high levels of acute malnutrition. Past events in South Sudan demonstrate the potential for new shocks to quickly arise and isolate households from food sources, leading to extreme hunger. An end to cycles of conflict and displacement, accompanied by multi-sectoral interventions to strengthen basic health and sanitation services, improve flood management infrastructure, and rebuild livelihoods and coping capacity, is required to end the risk of Famine (IPC Phase 5).

**Current food security outcomes, October 2022**



Source: FEWS NET

FEWS NET's classifications are [IPC-compatible](#). IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

### ***The risk of Famine (IPC Phase 5) is of greatest concern in Fangak and Canal/Pigi through mid-2023***

While Famine (IPC Phase 5) is not currently considered the most likely scenario in South Sudan during the outlook period through May 2023, there continues to be a credible alternative scenario in which Famine (IPC Phase 5) could occur. This risk is evident by sustained high levels of acute food insecurity and acute malnutrition, in addition to numerous historical examples in which conflict has isolated households and driven rapid deterioration in acute food insecurity outcomes. As a result, FEWS NET assesses that it remains plausible that a Famine (IPC Phase 5) could occur under certain conditions, particularly in areas where there are already large populations in Emergency (IPC Phase 4), and there is a potential for ongoing shocks to worsen.

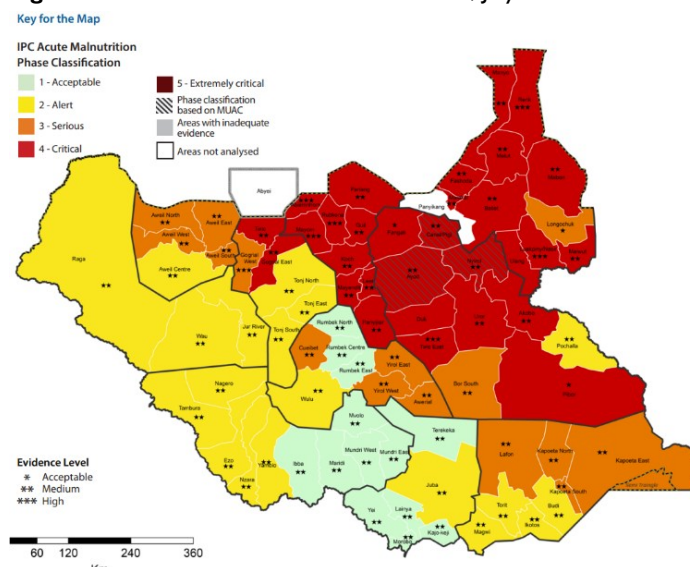
Prolonged conflict, consecutive years of widespread flooding, and macroeconomic challenges remain the key drivers of the country's widespread acute food insecurity. These shocks and their secondary impacts on crop and livestock production and staple food prices are driving low food availability and low food access across most of the country. In areas significantly affected by conflict and flooding, livestock ownership has markedly declined relative to 2018-2019, resulting in limited household income and access to food from milk and livestock sales. Crop production in 2022, which is estimated to be similar to or somewhat lower than last year, remains well below pre-conflict levels. In this context, many households remain heavily reliant on wild food sources (fishing, hunting, and gathering) and purchasing staple grains, even as income-generating opportunities remain insufficient and food prices outpace labor wage rates.

Available data on food consumption and livelihood coping used at the IPC analysis corroborates the very poor food security conditions, suggesting a large proportion of the population continues to face moderate or severe hunger and engage in negative and unsustainable coping in an effort to mitigate consumption deficits. Persistent food consumption gaps, exacerbated by poor access to clean water, sanitary conditions, and nutrition services, are driving high levels of acute malnutrition (Figure 1) among children, putting them at increased risk of death.

South Sudan will continue to face one of the world's worst food security emergencies as recurrent conflict, weather, and economic shocks limit food availability, regularly cut off household access to food, and limit households' capacity to recover from past shocks. While Famine (IPC Phase 5) is not considered the most likely outcome in the projection period, there is a credible, alternative scenario in which Famine (IPC Phase 5) could occur, given the [high proportion of the population already facing acute food insecurity](#) and the potential for the severity of ongoing shocks to increase. FEWS NET assesses that Fangak and Canal/Pigi counties in Jonglei state – where large swaths of land remain inundated and conflict is ongoing along the border with Upper Nile – are among the areas of greatest concern for this [risk of Famine \(IPC Phase 5\)](#). In these areas, if conflict were to escalate, restrict household movement, and isolate households in inaccessible areas such that humanitarians were unable to reach the worst-affected households for a prolonged period, Famine (IPC Phase 5) could occur.

It is critical to emphasize that Emergency (IPC Phase 4) outcomes reflect an already elevated level of hunger-related mortality, which begins to occur prior to reaching the [Famine \(IPC Phase 5\) criteria](#). A significant scale-up of multi-sectoral assistance is needed urgently in South Sudan in order to reduce acute malnutrition and associated mortality levels and end the risk of Famine (IPC Phase 5).

**Figure 1. IPC acute malnutrition classifications, July-October 2022**



Source: South Sudan IPC Technical Working Group

## NATIONAL OVERVIEW

### Current Situation

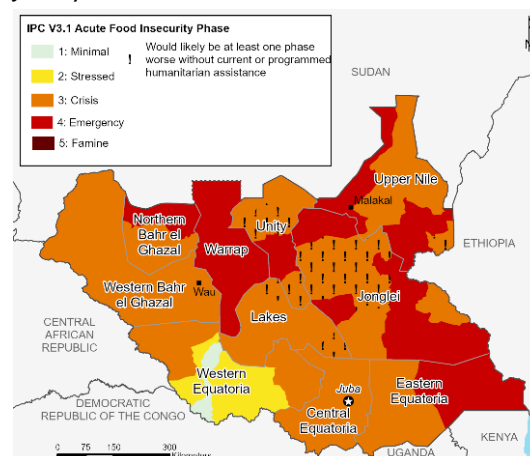
**Conflict and insecurity:** Despite some renewed progress towards key agreed-upon benchmarks of peace since the 2-year extension of the transitional government, including the training and graduation of the unified forces, passage of the constitution-making process bill, and the re-constitution of the electoral commission, deep rifts remain unreconciled around the country. Outbreaks of conflict continue to occur, interfering with household movement in search of livelihood options, trade flows and market functionality, as well as delivery of humanitarian food assistance to the most vulnerable, conflict- and flood-affected households.

Of the highest concern are the conflict-affected populations in northern Jonglei and south-western Upper Nile, where **over 40,000 people have been displaced since the start of the violence in mid-August**. The fighting, primarily between groups loyal to General Simon Gatwech's SPLA-IO Kitgwang and General Johnson Olony's Agwelek forces that has also inflamed divisions between Nuer and Shilluk communities, displaced over 13,000 people from New Fangak to Old Fangak and Kurwai in Canal/Pigi in late August and thousands more to the Malakal Protection of Civilians (POC) site in Kodok town in Fashoda county in mid-October. Amidst continued mass mobilization of armed Gawaar Nuer youth from Ayod and their concentration at Maat (Canal/Pigi), the likelihood of continued violent outbreaks remains high. The ongoing insecurity in the Greater Upper Nile has restricted WFP's ability to move lifesaving supplies by river from Bor to Malakal and created unpredictable access from Malakal to Kodok.

In other areas of Jonglei, cattle raiding attacks and road ambushes persist, with over **370 heads of cattle** raided in mid-October in parts of Uror and **Nyiröl** counties by armed youths from Pibor Administrative Area. Along the Abyei/Twic border in northern Warrap, renewed fighting between Twic and Ngok Dinka armed youth in mid-October displaced an **additional 1,000 people**. The South Sudanese People's Defense Forces (SSPDF) were deployed in the area, and a tentative **calm has endured in Abyei** since the Twic Mayardit community of Warrap State and the Ngok Dinka of Abyei agreed to cease hostilities and engage in a peace dialogue. However, trade flows and market functionality have yet to recover, interrupted by the continued posturing of armed youth. In other parts of Warrap, **16 youths were killed**, eight others wounded, and an unknown number of cattle raided in mid-October at Panyan-Bech in Tonj East county by youths suspected to be from the Luany-Jang section.

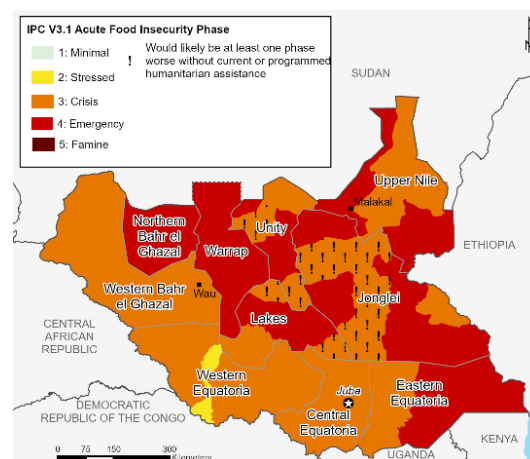
Outside of these conflict hotspots, outbursts of violence continue to flare in parts of central and northern Unity, Greater Tonj and Twic of Warrap, and parts of Central and Eastern Equatoria despite community-level peace initiatives, which reflect to some extent the entrenched nature of inter-and intra-communal clashes, revenge killings, ambushes and farmer-herder conflict. In Panyume *payam* of Morobo county in Central Equatoria, soldiers belonging to Machar's SPLA-IO reportedly carried out violent robberies and assaults against civilians, displacing over 1,000 people to their headquarters in late September. Armed clashes involving NAS, SPLA-IO and SSPDF in parts of rural Yei, Lainya, and Morobo are threatening second season harvesting and trade flows to Yei markets. In Eastern Equatoria, road ambushes along Torit-Kapoeta, Camp 15-Chukudum, and the Ikotos-Tsertenya roads continue to disrupt trade flows and the movement of traders and humanitarian aid workers. In Western Equatoria, rising tensions among the SPLA-IO regarding participation in the ongoing unification of forces could lead to a return of conflict in the State.

Projected food security outcomes, October 2022 to January 2023



Source: FEWS NET

Projected food security outcomes, February to May 2023



Source: FEWS NET

FEWS NET's classifications are IPC-compatible. IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

**Rainfall performance:** Although cumulative rainfall during the June to September main rainfall season was generally higher than normal for most of north-eastern and north-western South Sudan, the north-central and southern parts of the country registered overall rainfall deficits of between 5 and 30 percent for this period (Figure 2). Drier-than-average conditions and erratic rainfall distribution in the Equatorias and parts of Lakes and Unity from the start of the season through mid-July negatively affected crop development in some areas.

Meanwhile, wetter-than-average conditions in Northern and Western Bahr-el-Ghazal, Upper Nile and central-eastern Jonglei, compounded by the poor recession of last year's flood waters and elevated flow from Lake Albert in Uganda, have contributed to increased river water levels and record flood extents along the Bahr-el-Ghazal and Nile Rivers. Furthermore, above-average rainfall continued through October and into early November in many parts of the country. The exceptions included Boma in Greater Pibor Administrative Area, Kapoeta East of Eastern Equatoria, north-western Raga of Western Bahr el Ghazal, and large swathes of Western Equatoria, where rainfall deficits of between 5 and 30 percent in October signaled the timely cessation of the rains in these areas.

**Flooding:** This year's flooding marks the fourth consecutive year of abnormally high flooding in the country, which reached record extents in some areas and is [estimated to have affected over 1 million people](#) in 36 counties and over 20,000 people in the southern part of the Abyei Administrative Area as of the end of October. The most severely flood-affected areas include Northern Bahr el Ghazal, Warrap, Upper Nile, and Unity, followed by Western Equatoria, Lakes, and Jonglei. By comparison, flooding affected the highest number of people in eastern Jonglei and Upper Nile in 2019, in Pibor in 2020, and in northern Jonglei, central and northern Unity in 2021 (Figure 3).

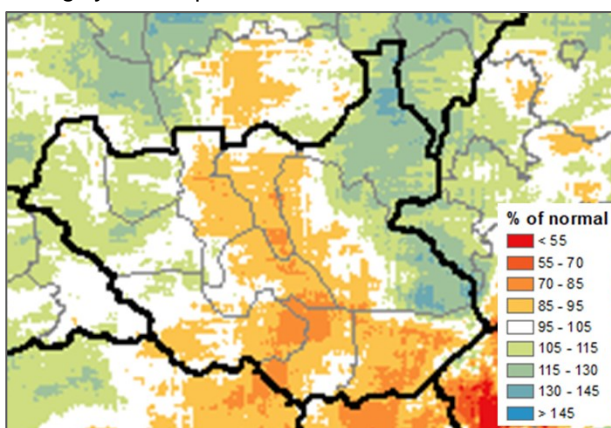
In flooded areas, significant damage and destruction was reported for household property, crops, and livestock, as well as to key infrastructure such as roads, bridges, dykes, markets, and education and health facilities.

The resulting disruption of services and contamination of water supplies also increased the risk and spread of waterborne diseases, likely contributing to a rising burden of malnutrition. At the same time, nutrition cluster partners reported that the floods and insecurity have affected operations at a total of 111 nutrition sites and hindered the delivery of nutrition services to 108,000 children and pregnant and lactating women (PLW) in 42 counties in eight states, including Jonglei, Unity, Upper Nile, Lakes, Warrap, Western Bahr el Ghazal, Northern Bahr el Ghazal and Western Equatoria. In addition, the ongoing flood response is hampered by renewed violence and insecurity, inaccessibility due to impassable roads, broken bridges, flooded airstrips, the lack of air assets, the lack of critical core pipeline supplies, and funding constraints.

**Crop production and harvest:** The harvesting of main season sorghum is ongoing in unimodal and bi-modal areas of South Sudan. In the unimodal zone, harvesting of groundnuts as well as both late and early maturing sorghums crops was completed in the Greater Bahr el Ghazal and Upper Nile regions. In Cueibet, Greater Rumbek, and Greater Yirol in Lakes State, harvesting of long maturing crops and groundnuts is underway. In bimodal Greater Equatoria region, second season crops, particularly maize, are at late vegetative stage while groundnut, sesame, and long maturing sorghum are at flowering stages, including ratoon sorghum crops in the Greater Kapoeta area.

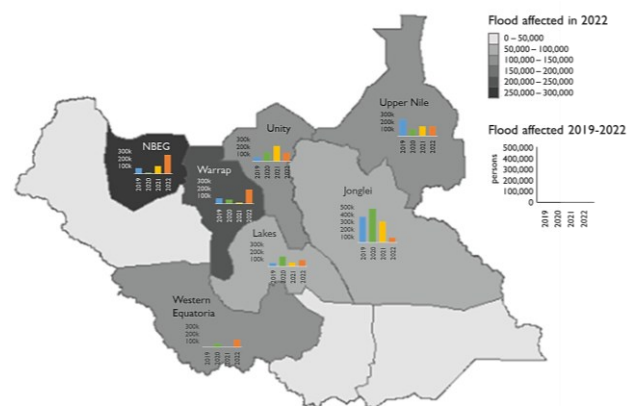
Although official Crop and Food Security Assessment Mission estimates are not yet available, many households across South Sudan report facing crop production challenges in 2022, including: prolonged dry-spells and moisture deficits in north-central

**Figure 2.** Cumulative rainfall as a percent of the 1981-2020 average, June to September 2022



Source: USGS

**Figure 3.** Flood-affected populations by state in South Sudan as of October 31, 2022, with trends from 2019 to 2022



Source: consolidated from OCHA reports (2019-2022)



and southern South Sudan; flood-related disruption to household planting and significant damages to crop fields in flood plains and in the Sudd region; conflict-related impacts on access to fields across large parts of conflict-affected Panyikang, New Fangak, Canal/Pigi, Malakal, and Fashoda counties, particularly following surges in conflict in mid-August and October; and economic constraints from factors such as high commodity prices and reduced purchasing power that is hindering households' access to inputs. As such, the performance of main season harvests likely varies between states, but total national production is expected to be similar to, or lower than, last year and the five-year average. This is corroborated by FAO's preliminary main season harvest estimates, which indicate that the main harvest is likely lower than last year's in flood- and conflict-affected states including much of Greater Upper Nile and Greater Bahr el Ghazal, while the overall second season harvest is likely to be higher than last year's in relatively calm areas in bimodal Greater Equatoria.

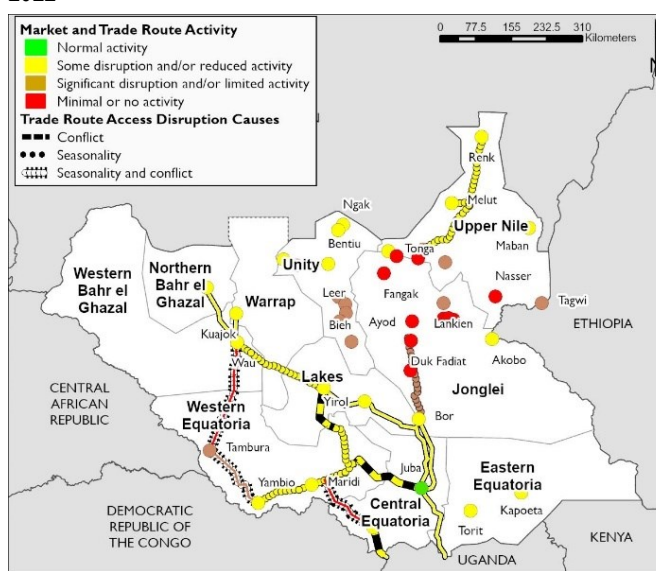
**Livestock production:** Livestock production systems continue to struggle under the impacts of prolonged dry spells, flooding, and/or conflict across most of the pastoral and agropastoral livelihood zones of South Sudan. Prolonged dry spells in June and July in central-southern Lakes, parts of Mundri, eastern Pibor, and the Kapoetas led to severe water and pasture shortages for livestock, negatively affecting livestock body conditions. In contrast, in the severely flood-affected areas, poor livestock body conditions have been due to damaged or submerged pastures, water contamination, high waterborne disease incidence, and flood-imposed livestock movement restrictions. Key informants report increased incidence of waterborne diseases such as Black Quarter (BQ), liver fluke, and Foot and Mouth Diseases (FMD), especially in the most severely flood-affected areas of Aweil East, West, and North of Northern Bahr el Ghazal, Gogrial, Twic, and greater Tonj of Warrap, and Duk of Jonglei. Overall, livestock body conditions in these areas affected by shocks are seasonally poor to fair, while by contrast, livestock body conditions in non-affected areas are fair to good due to sustained access to water, pasture, and better grazing resources.

In Greater Bahr el Ghazal, southern Abyei Administrative Area, Upper Nile regions and Western Equatoria state, significant livestock displacement caused by both conflict and flood events has led to the concentration of herds in high ground grazing areas and increased incidence of livestock diseases and deaths, compounded by limited access to veterinary services. The loss of livestock to disease and poor health, conflict and cattle raiding, and distressed sales continues to depress livestock ownership rates, particularly in areas dealing with near-decimated levels of livestock ownership, such as Fangak, Canal/Pigi, Pibor, Leer, and Mayendit counties. The decline in livestock ownership has further negatively affected household incomes from, and access to, livestock products.

**Market and trade:** Although markets in the state capitals are operating normally, markets in the rural areas of Western and Northern Bahr el Ghazal, Unity, Jonglei, Upper Nile and some part of Equatoria regions remain significantly disrupted due to the flooding of feeder roads (Figure 4). Additionally, conflict and insecurity have disrupted market functioning and market supplies in Mayom of Unity; Fangak and Canal/Pigi of Jonglei; Malakal, Fashoda, Panyikang, and Kodok town of Fashoda in Upper Nile; and Twic of Warrap.

Cross-border trade continued through the main border crossings in the Equatorias and Greater Bahr el Ghazal, but overall supply levels have been lower than last year due to the below-average harvest and high regional demand. According to the [East Africa cross-border trade bulletin](#), the import volumes of maize and sorghum from Uganda during the third quarter of 2022 (July-September) was 64-65 percent, 75-79 percent, and 25-43 percent lower than the second quarter (April-June 2022), the same period last year, and the five-year average, respectively. Imports of sorghum from Sudan were also 47 percent lower in the third quarter compared to the second, although slightly higher than the same time last year (+9 percent) and nearly double the five-year average (+96 percent) due to reduced inflows from Uganda.

**Figure 4.** Market functioning and trade route activity, October 2022

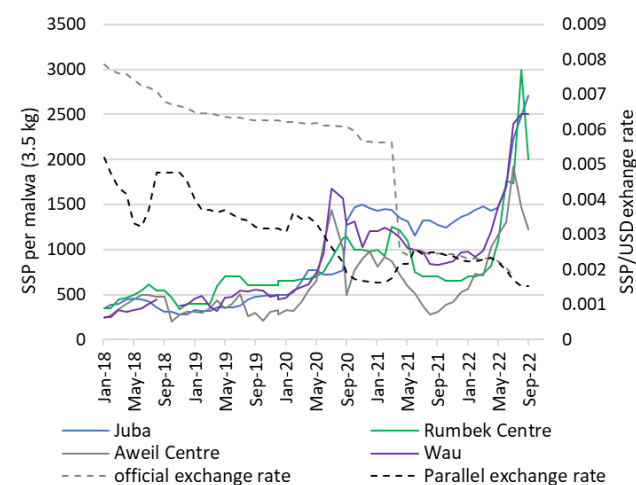


Source: FEWS NET

**Staple food prices:** Retail prices of the primary grains (maize and sorghum) have started to stabilize or decline in recent months (September and October) due to increased availability of food stocks from the main season harvest, but are still on average higher than three months ago and considerably higher than the same time last year and the five-year average (Figure 5). According to price data monitored through CLiMIS, the price of white sorghum per *malwa* (3.5 kg) was around 120-290 percent above last year in Rumbek Centre, Aweil Centre, Wau, and Juba, and 200-330 percent higher than the five-year average. This longer-term trend of price increases reflects international price trends as well as high operational costs related to insecurity and poor road infrastructure, in addition to low local production and depreciation of the local currency. In rural markets where the harvest is low or minimal, prices have remained extremely high. High staple food prices are limiting household financial access to food across the country.

The terms of trade (ToT) of daily casual labor to one *malwa* of white sorghum have remained similar to September and same time last year in Juba and Aweil, respectively. Sorghum prices declined in October with the harvest of quick-maturing sorghum, yet wages remained atypically low due to fewer income-earning opportunities amid low economic activity.

**Figure 5.** Five-year trends in exchange rate and the price of a *malwa* (3.5 kg) of sorghum in four key markets



Source: Data from CLiMIS

**Humanitarian food assistance:** Despite the ongoing harvest, humanitarian food assistance needs remain high across the country, driven by the significant flood- and conflict- induced crop and livestock losses, as well as disruptions to trade and market functioning. WFP's September 2022 food assistance distribution report confirmed food assistance reached 1.13 million people with 12,162 MT of food through the general food distribution (GFD) and food for assets programs, representing 9.1 percent of country population and roughly 14.3 percent of the total estimated population in need. Based on WFP's bi-weekly food distribution update from October 30, distributions of GFD for 18 counties under supplemental funding were completed in Aweil South, Aweil West, Rumbek East, Rumbek Centre, Gogrial East, Longochuk, and Aweil North, but were delayed in Yirol East and Yirol West due to contracting issues. Distributions were ongoing in Jikmir of Nassir and Fashoda in Upper Nile. For Akobo East, food deliveries were distributed from Gambella. Additionally, food distributions under the rapid response mechanism (RRM) were completed in Akobo West and Uror of Jonglei; Maiwut of Upper Nile; and Leer and Panyijiar of Unity as of the end of the month, but were still ongoing in Nyirol and Ayod of Jonglei and Mayendit and Koch of Unity. Distributions to flood-affected IDPs in Mayom, and Bentiuf of Rubkona were also ongoing, targeting 7,000 and 136,000 people, respectively. Finally, in Canal/Pigi, delivery of food remained challenging due to insecurity and limited air operations. As such, food distributions remained suspended in Atar 3 and Tonga due to insecurity and conflict.

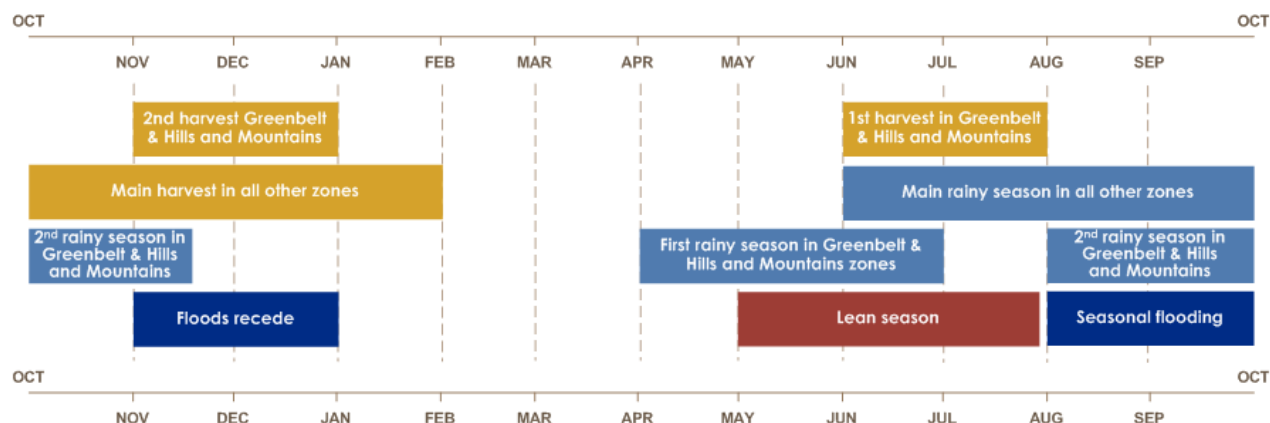
### Current Food Security Outcomes

Multiple shocks – including protracted conflict, consecutive years of flooding and a macroeconomic crisis characterized by currency depreciation and rising prices – continue to drive high levels of acute food insecurity in South Sudan. In this context, the [national IPC acute food insecurity and acute malnutrition analyses](#) conducted from October 3-15 estimated 6.6 million people were still experiencing high levels of food insecurity and 1.4 million children were still acutely malnourished in October-November harvesting period, reflecting that scale of need continues to outpace food and nutrition assistance deliveries. These estimates do represent a modest improvement in food security compared to the [March 2022 IPC acute analysis](#), driven by sustained relative calm in Western Equatoria and Western Bahr el Ghazal; however, food security deteriorated notably in some counties of greatest concern. Populations are classified in Catastrophe (IPC Phase 5) in Fangak, Canal/Pigi, and Akobo counties of Jonglei State, plus Pibor county in the Greater Pibor Administrative Area.

The most food insecure states where more than half of their populations are facing Crisis (IPC Phase 3) or worse acute food insecurity in the presence of assistance include Jonglei, Unity, Northern Bahr el Ghazal, Upper Nile, Warrap, and Lakes. At the county level, 35 of the 79 counties are classified in Emergency (IPC Phase 4), 42 in Crisis (IPC Phase 3), one in Stressed (IPC Phase 2), and one in Minimal (IPC Phase 1). The counties of high concern are those with a significant proportion of their populations estimated to be in Emergency (IPC Phase 4) or worse, which includes: Leer, Mayendit, and Rubkona of Unity; and

Akobo, Ayod, Canal/Pigi, Fangak, Nyirol, and Pibor of Jonglei. Other counties of concern include Gogrial East, Twic, Tonj North, Tonj East, and Tonj South of Warrap; Aweil North and Aweil East of Northern Bahr el Ghazal; Longochuk, Nasir, and Malakal of Upper Nile; and Uror and Twic East of Jonglei. The most severe acute food insecurity is in locations with persistent risk to acute food insecurity that has been worsened by frequent climate-related shocks (severe flooding and dry spells), the macro-economic crisis, and conflict and insecurity. Urgent scaling-up of multi-sectoral humanitarian assistance to save lives and prevent the collapse of livelihoods is needed.

## SEASONAL CALENDAR FOR A TYPICAL YEAR



Source: FEWS NET

### Assumptions

The most likely scenario from October 2022 to May 2023 is based on the following national-level assumptions:

- Conflict and insecurity:** Despite the re-commitment of the national government to the peace deal through the graduation of the Necessary Unified Forces (NUF) in late August and the likely deployment of security forces in some areas, conflict and insecurity are expected to continue to threaten lives and livelihoods.
  - Ongoing violence in northern Jonglei and south-western Upper Nile suggests conflict will likely continue and will periodically displace households and cause the destruction of property and evacuation of humanitarian aid workers. Reduced conflict is likely through November as flooding events from the end of the rainy season limit movements. However, conflict will likely then increase through May with the dry season.
  - Clashes between government forces and the SSPM/A under General Buay in northern Unity will likely continue due to ongoing tensions following the extrajudicial killings of officers who were suspected of killing the Mayom County Commissioner and ten others in July. Conflict in Unity is likely to match that of 2021, following seasonal trends with relatively higher levels of conflict occurring between December and May.
  - In the Greater Tonj of Warrap, inter-communal violence and insecurity are expected to increase as livestock keepers concentrate in the dry season grazing areas, increasing the incidence of raids, counterraid, and clashes between herders and farmers. However, conflict levels are likely to be lower than last year due to the deployment and likely additional deployment of security forces in parts of Greater Tonj.
  - In Eastern, Central, and parts of Western Equatoria, clashes between herders and farmers are likely to persist. Furthermore, armed confrontations are likely to occur in several areas despite the graduation of NUF in Juba, Torit, and Maridi in August/September, because of delays in the deployment of security forces and failure to integrate all armed opposition groups – most notably the National Salvation Army. Violence against civilians will likely follow the seasonal pattern of increasing through May, meeting levels observed in 2022.
- Rainfall and flooding:** Based on the NMME and WMO forecasts, the July-November 2022 second rainy season in bimodal South Sudan is most likely to conclude at near-average levels in most areas, with localized rainfall deficits in south-eastern Greater Equatoria following low rainfall performance in August. Persistent wet conditions are still likely to delay the recession of floodwaters in the Sudd Wetland and major tributaries. Atypically high flood extent will likely persist well

into 2023. The March to May 2023 first-season rains are forecasted to be average, though there is greater uncertainty in this forecast given its long-range nature.

- **Harvest:** The September-December main season harvest, which coincides with second-season production in bimodal areas, is expected to be similar to or somewhat lower than last year and the five-year average at the national level. The harvest will vary between states depending on localized rainfall performance, insecurity, and farmer-herder conflicts. The second season harvest is likely to be higher than last year's in bimodal Greater Equatoria due to the relative calm, but below average in significantly flood- and conflict-affected areas of Greater Upper Nile and Greater Bahr el Ghazal.
- **Livestock:** While at the national level livestock ownership rates will remain similar to 2022 going into 2023, there will continue to be considerable variation across South Sudan depending on the level of exposure to flood and conflict shocks. Available survey data used for the October 2022 IPC analysis suggested livestock ownership declined notably from last year in most of Greater Upper Nile, while it has increased in wide areas of Greater Bahr el Ghazal and Greater Equatoria and parts of eastern Upper Nile. These trends are likely to be sustained into 2023. In flood-affected parts of Unity, Jonglei, Upper Nile, parts of Warrap, Northern Bahr el Ghazal, and Lakes, livestock production will likely be poor due to widespread livestock losses (floods and cattle raids), water contamination, pasture loss, and rising waterborne diseases. As such, household access to milk and meat will also be lower than normal in these states.
- **Fish and wild foods:** The availability of fish and wild foods will remain high through February and decline through April/May as flood waters recede during the dry season. However, access to these natural food sources will vary depending on the level of floodwaters, as well as on security-related mobility restrictions, particularly in the most severely conflict- and flood-affected counties in the Greater Upper Nile and parts of greater Bahr el Ghazal. In Western Bahr el Ghazal and parts of Greater Equatoria, households' access to these food sources will likely be average.
- **Macro-economy:** South Sudan's economy is expected to remain poor, due in large part to the low oil production around 160,000 bpd, driven by flooding events in the oil fields, production quotas imposed by OPEC, and the continued disruption of global supply chains. Revenues from oil exports are likely to fluctuate, limiting government revenue and driving continued depreciation of the SSP. The SSP is expected to depreciate to at or above 600 SSP/USD, driving a general increase in the price of most food and non-food items and reducing household purchasing power amid stagnant wages.
- **Labor demand and wages:** The availability of labor opportunities will vary across urban centers but remain limited by poor macroeconomic conditions. Based on the observed wage rate trends in CLIMIS, wage rates are unlikely to keep pace with staple food price increases, driving lowering terms of trade. Rural wage labor opportunities will decline through the projection period as the agricultural season has ended.
- **Cross-border trade:** Based on FEWS NET's monthly cross-border trade monitoring data and observations of low harvests in exporting countries in the region combined with high transportation costs, imports are expected to decline from Uganda and Sudan when compared to last year, particularly in markets outside of Juba. As a result, imports from Uganda and Sudan during the fourth quarter of 2022 and the first quarter of 2023, will remain similar to the third quarter of 2022.
- **Staple food prices:** Based on FEWS NET's integrated price projections and likely supply and demand factors, the retail price of a *malwa* (3.5 kg) of white sorghum is expected to trend 35-335 percent higher than last year and 135-320 percent above the five-year average in Juba, Bor South, Wau, and Aweil markets due to high market dependency, low import levels, low harvest, and high supply costs. The price per *malwa* is projected to range from 1,200 to 2,810 SSP. In absolute terms, the highest price is expected in Bor South and the lowest price in Aweil Center.
- **Household purchasing power:** Nationally, household purchasing power will remain below average due to the continued poor macroeconomic conditions and high staple food prices. As such, household financial access to food will remain low.
- **Humanitarian food assistance:** Based on WFP's humanitarian food assistance plans made available during the October IPC, WFP plans to reach 17 percent of the national population with a 50 percent ration from December through March, and scale-up to reach 18 percent of the country population from April through May. Based on the past trends and funding shortfalls, WFP is likely to prioritize food assistance deliveries in areas of extreme concern with significant populations in Emergency (IPC Phase 4) or worse, though deliveries will likely face insecurity and funding-related disruptions.



### Most Likely Food Security Outcomes

Over the length of the full harvesting period stretching **from October 2022 through January 2023**, acute food insecurity will improve slightly with the harvest, although such improvements will vary widely across the country and be negligible to minimal in areas where flooding and/or conflict has significantly impacted crop and livestock production. As such, the proportion of households facing large to extreme food consumption gaps and with high dependence on negative coping indicative of Emergency (IPC Phase 4) or worse will decline from October/November to December 2022 to March 2023 period. The largest improvements are expected in Western Bahr el Ghazal, Western Equatoria, and Central Equatoria. However, in flood- and conflict-affected areas in Northern Bahr el Ghazal, Jonglei, Upper Nile, Unity, Warrap, and Lakes, many households will continue to face limited access to own-produced food, restricted physical access to markets due to flooding and conflict, and constrained financial access to food due to declining purchasing power, and are thus expected to depend more heavily on natural food sources, including fish and wild foods, and humanitarian food assistance. In addition, Catastrophe (IPC Phase 5) will likely persist in Fangak, Canal/Pigi, and Akobo of Jonglei and Pibor County in the Greater Pibor Administrative Area, with populations in Catastrophe (IPC Phase 5) in these counties. During this time, food assistance is expected to mitigate the severity of acute food insecurity in 11 counties in Upper Nile, Unity, Jonglei, and Lakes, leading to Crisis! (IPC Phase 3!). However, Emergency (IPC Phase 4) outcomes will likely persist in 26 counties.

**From February to May 2023**, which overlaps with the post-harvest period and the typical start of the lean season period, food stocks at the household level will begin to deplete, increasing dependence on markets at a time when food prices are expected to be higher. Purchasing power to be further constrained amidst the deteriorating macro-economic context and declining wage opportunities, and household physical access to markets to be limited, particularly in flood- and conflict-affected areas. As the availability of wild foods is also expected to decline during the dry season, households are likely to turn increasingly to coping strategies such as begging or movement to nearby villages to access food or income. It is estimated that 41 counties will be in Emergency (IPC Phase 4), concentrated in Jonglei, Unity, Northern Bahr el Ghazal, Upper Nile, Warrap, and Lakes. Food assistance needs among the population will most likely increase and continue to exceed currently funded levels of humanitarian assistance. Populations in Catastrophe (IPC Phase 5) are expected in Fangak, Canal/Pigi, Akobo of Jonglei, Pibor County in the Greater Pibor Administrative Area, and Leer and Mayendit of Unity. During this time, food assistance is expected to mitigate the severity of acute food insecurity in 9 counties to Crisis! (IPC Phase 3!).

### Events that Might Change the Outlook

**Table I.** Possible events over the next eight months that could change the most-likely scenario

| Area            | Events                                                                                                                                                                                                            | Impact on food security outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>National</i> | A significant conflict or weather event that isolates households from moving towards food and income sources for a prolonged period; humanitarian access is also limited.                                         | In the event that a conflict or weather shock restricts household movement, then worst-affected households would face significant difficulty accessing food from crops, livestock, markets, and/or wild foods. Limited humanitarian access would also restrict the capacity to respond with assistance. Such events have occurred numerous times in recent years in South Sudan and have repeatedly highlighted that food security can deteriorate rapidly in these conditions. If such an event were prolonged, Famine (IPC Phase 5) would be likely. |
| <i>National</i> | A breakdown in the implementation of the remaining key provisions of the peace deal, and continued exclusion of hold-out armed opposition group, leading to more extreme, widespread conflicts and displacements. | A rise in conflict and insecurity events would displace more households and have an even more significant negative impact on crop production, trade flows, and market functioning. As a result, Emergency (IPC Phase 4) outcomes will be even more widespread than assumed, driving up the proportion of population in Catastrophe (IPC Phase 5).                                                                                                                                                                                                      |

| Area                                                                   | Events                                                                                                                                                   | Impact on food security outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>National</i>                                                        | A further deterioration in macro-economy, low levels oil production and exports, and a rise in staple food prices beyond already anticipated high levels | If the macro-economy deteriorates further than already anticipated, marked by spiking food prices, limited income-earning opportunities that severely restrict household capacity to purchase food, acute food insecurity will worsen. This will also be driven by a likely corollary decline in economic activities and collapse of scale-scale businesses, leaving many households with even fewer income-earning opportunities and a lower ability to purchase food. These declines would likely outpace the positive impacts of humanitarian assistance and the harvest, and Emergency (IPC Phase 4) outcomes would be more widespread.                          |
| <i>Fangak and Canal/Pigi, Jonglei</i>                                  | Cessation of conflict and violence by SPLA-IO Kitgwang forces and General Olony's Agwelek forces, permitting relative calm within the communities        | If the government reaches an agreement with the armed groups loyal to General Simon's SPLA-IO Kitgwang forces and General Johnson Olony's Agwelek forces and there ensues a subsequent cessation of violence and restoration of relative calm in the community, then there would be a resumption of trade flows, market recovery, and unhindered delivery of humanitarian food assistance. Additionally, household movement in search of natural food sources such as fish and wild foods would also improve significantly, enabling reach to distant areas. As such, the size of the county populations facing Crisis (IPC Phase 3) or worse outcomes would reduce. |
| <i>Akobo and Pibor County in the Greater Pibor Administrative Area</i> | A rise in intercommunal conflict, violence, and cattle raids to levels similar to 2020 or higher                                                         | An escalation in inter-communal conflict and violence and cattle raids would displace households and significantly disrupt harvesting activities, trade flows, and market functioning. Additionally, the rise in conflict/insecurity would limit household access to natural food sources and hinder the delivery of food and non-food assistance. Emergency (IPC Phase 4) outcomes, with a rise in the share of the population in Catastrophe (IPC Phase 5).                                                                                                                                                                                                        |
| <i>Leer and Mayendit of Unity</i>                                      | Lower levels of conflict and insecurity and floodwaters substantially recede to normal levels                                                            | If intercommunal conflict and related insecurity were to reduce further, and floodwaters substantially recede in both areas, this would permit freer household movement, improve trade and market functioning, and increase humanitarian access in both areas. Food consumption gaps would reduce, leading to less extreme acute food insecurity.                                                                                                                                                                                                                                                                                                                    |

## AREAS OF CONCERN

***Leer and Mayendit of Unity State: Nile Basin Fishing and Agropastoral zone (SS08) and Northwestern Flood Plain Sorghum and Cattle zone (SS07) (Figure 6)***

### *Current Situation*

**Flooding and displacement:** Successive years of severe flooding has resulted in high levels of household displacements, related disruption to household engagement in crop and livestock production, trade and market functioning, and asset loss in Leer and Mayendit of central Unity, which is driving extremely high household risk of acute food insecurity in both areas. Of great concern are households in villages such as [Juong, Payak, Thonyor, Muom, Din-din, and Adok of Leer](#); and [Dablual, Kuerdung, and Rubnor of Mayendit](#) where an estimated 24 and 62 percent of county population respectively were displaced or affected by ongoing flooding in 2022, according to the Interagency Rapid need Assessments (IRNA) conducted in September 2022. Key informants reported that the scale of the 2022 flooding surpassed that of 2021, even submerging some highlands in Rubkuay, Rubnor, Kuerduong, Maper, Bathlar, Mirnyal, Pabuong, Padeah, Dhorgapni, Thowkuok, Gam, Luom, Tutnyang, and Dhorjak, which were not flooded in 2021. This brings the total number of flood-affected to approximately 85 percent of the population. Furthermore, both key informant information and FAO flood extent estimates indicate nearly

50-75 percent of the areas in Leer and Mayendit were inundated, affecting boreholes and latrine facilities, with a resultant high prevalence of waterborne diseases and water pollution.

**Conflict:** Although the level of both intercommunal violence and cattle raiding has significantly reduced in recent months relative to observed levels in the first half of 2022, due in part to movement restriction caused by high floodwater, the lingering impacts of years of protracted conflict, asset losses, security restrictions, and multiple household displacements have further exacerbated household vulnerabilities and significantly weakened traditional livelihood systems and resource sharing mechanisms. Despite the prevailing relative calm in the areas, reports of tension and defections in response to increasing marginalization and threats of livestock confiscation by government-allied youth militia in an effort to consolidate government position and fragment the opposition highlight the potential for renewed conflict in central Unity. Both key informant information and WFP bi-weekly reports confirm that the tension is instilling fear, restricting household movement, and interfering with humanitarian deliveries and services.

**Seasonal performance and crop production:** Although the June to September main rainfall was generally timely, total cumulative rainfall was lower than that of the same period in 2021 and of the long-term average for both areas. Household planting and cultivation were significantly limited or disrupted by conflict, household displacements, and high residual floodwater. Although key informants reported some planting occurred in highland areas of Mayendit, most of the crop fields were inundated and destroyed by floodwater. In Leer, very limited planting occurred due to inter-communal conflict and insecurity that displaced more than half of the population to Leer town, the military training center, and islands/swamps for protection. This is consistent with available survey data from the October 2022 IPC analysis suggesting low levels of planting for the second year in a row, compared to previous years in which high proportions of households planted in both counties. In the context of reported low planting and the severity of the 2022 floods on crop production and harvests, the 2022 harvest is likely to be lower than last year and the five-year average.

**Livestock production:** Livestock production was typically a key food and income source in both livelihood zones; however, livestock contribution to the normal traditional production system has been significantly threatened by years of protracted intercommunal conflict and cattle raiding, as well as diseases and flood-related livestock mortality. Livestock ownership remains very low according to recent household survey data from humanitarian partners and key informant reports, and is likely lowest among the poorer households who are unable to rebuild herds amidst the ongoing economic, climatic, and conflict shocks. Field monitoring reports and key informant information gathered from both areas in October also indicate livestock body conditions and production are generally poor and low due to the impact of floodwater on pasture and insecurity-related restrictions to livestock movement. As such, access to livestock products such as milk is likely to have remained relatively low in both areas of concern.

**Access to fish and wild foods:** Availability and access to fish and wild foods are typically high between October and February, and households are increasingly reliant on these food sources to mitigate food consumption gaps in the context of a decline in own-produced foods. While key informants report that relative improvement in the security situation in recent months is permitting a relative increase in household access to fish, wild foods, and bush products, household ownership of or access to fishing equipment remains limited, and dangerously high floodwaters and insecurity continue to limit these activities in general.

**Trade and market system:** Typically, during this period, food prices tend to decrease due to increased supply from harvests and reduced demand from households given their own-produced stocks. However, available survey data and key informant information suggests that household dependence has atypically increased in these areas due to prevailing limitations on households' ability to produce their own food. Concurrently, poor market functionality persists. Domestic trade flow and market supplies from Bentiu and Juba via the Nile River are significantly disrupted by flooding and insecurity. The few functional markets, such as Rubkuay in Mayendit and Leer town, have limited cereal stock due to poor feeder road conditions

**Figure 6.** Area of concern reference map: Leer and Mayendit of Unity state



and insecurity along trade routes.

**Staple food prices:** Analysis of available field monitoring price data in both areas confirms that the retail price per *malwa* (3.5kgs) of red sorghum is 95-127 percent higher in October than the same time last year due to high supply costs associated with flooding, insecurity, low stocks, and high fuel costs. Compared to the previous month (September 2022), the price of red sorghum was 138 percent higher in Mayendit, reflecting declining stocks and high transport costs, though the price remains similar in Leer due to its proximity to Adok port. The October 2022 price of medium-sized-goat remains similar to that observed in September in Leer and Mayendit, although 5-16 percent lower than the same period last year due to low effective demand. Based on the analysis of goat-to-sorghum ToT, households in Mayendit could purchase only 3.6 kg of sorghum with the sale of a goat compared to 8.1 kg in September 2022 and 8.6 kg in the same period last year, reflecting a decline of more than 50 percent from both periods. In Leer, livestock-owning households could purchase 5 kg of sorghum with the sale of medium-sized goat in October, which is roughly similar to September 2022, but about 5 kg lower (-52 percent) than the same period last year. This is likely limiting livestock-owning households' financial access to food.

**Humanitarian food assistance:** Humanitarian food assistance remains a key source of food for most households in these areas of concern, where primary food and income sources have been significantly reduced by the combined impacts of successive years of flooding and armed violence. However, assistance deliveries to these areas continue to face logistical challenges related to high floodwater and movement restrictions by insecurity. According to an analysis of WFP distribution reports from July to September, nearly 15 percent of the population in Mayendit and 40 percent of the population in Leer, on average, received more than a full ration monthly. At the end of October, WFP confirmed all planned distributions were complete in Leer and in four locations in Mayendit, despite the limited capacity for air drops. Nonetheless, some isolated households displaced to remote locations have limited access to assistance.

Given increased household exposure to multiple hazards and resultant large-scale displacements, crop and livestock losses, disruptions to trade and market functioning, limited household movement, and restrictions to humanitarian operations, a significant proportion of households in both areas of concern are facing severe acute food insecurity. Although sustained humanitarian food assistance through October is reaching a significant proportion of the population, it is assessed that the scale of need exceeds humanitarian reach. This is reflected in a high percentage of the population reporting engaging in severe coping and a moderate increase in the population reporting severe hunger, as assessed at the October 2022 IPC analysis, in the presence of ongoing assistance. Furthermore, the IPC analysis indicated Critical (15-29.9 percent) levels of GAM in both areas during the harvesting period. Based on the convergence of food consumption and livelihood coping, further supported by nutrition outcomes, Emergency (IPC Phase 4) outcomes are likely in both areas during the October harvesting period.

### Assumptions

In addition to the national-level assumptions, the following assumptions apply to this area of concern:

- Based on the IRI weekly rainfall anomaly and NMME and WMO forecasts, rainfall is expected to subside in November. However, due to above-average river flow, high soil saturation, and end-of-season rains, floodwaters in the Sudd Wetland and its major tributaries in the north are likely to persist into the dry season. This is expected to prevent displaced households from returning to their areas of origin and disrupt assistance deliveries, trade, and market functionality.
- Although intercommunal violence and cattle raiding incidences have both reduced in the third quarter relative to the first and second quarters of 2022, analysis of past trends suggests that ongoing tensions are likely to trigger localized violence and insecurity during the dry season as the flood waters recede and permit better, likely driving further displacement and disruptions to trade, market functioning, and assistance deliveries.
- Given the level of disruptions to trade and market functioning by high floodwater, low local production, insecurity, and illicit taxes along the trade routes in the areas, trade flows, and market supply are expected to be very limited through February, but improve slightly with floodwater recession through May. In combination with high global food and fuel prices, the retail prices of staple foods are expected to be higher than last year and the five-year average.
- Given the loss of livestock, security- and flood-related access restrictions to grazing areas, and expectations of continued inter-communal cattle raiding, livestock production and holding sizes are expected to reduce further. Household access to livestock products and incomes from livestock sales are likely to be limited. Among poor households, specifically,



holdings are expected to consist of a few goats and sheep but no cattle, resulting in overall low levels of milk access.

- Based on WFP's HFA plan made available during October IPC acute analysis, the organization plans to reach 35-55 percent of the population monthly in Mayendit and Leer with a ration meeting 67 percent of needs through November. Between December and May, the ration size will decline to roughly 50 percent. Overall, high floodwaters and insecurity will likely continue to pose major logistical challenges and limit the regular provision of assistance.

### *Most Likely Food Security Outcomes*

**From October 2022 through January 2023 harvesting period,** availability and access to all typical food and income sources are expected to be below to significantly below average in both areas of concern due to low crop and livestock production, the persistence of high residual floodwater, security restrictions to household and humanitarian movements, limited recovery in market and trade operations, and high staple food prices. As a result, a significant proportion of poor displaced households who have either no harvest or livestock to rely on, limited physical and financial access to markets, or restricted access to fish, wild leaves, livestock products, and food aid, are expected to continue facing large food consumption gaps, indicative of Emergency (IPC Phase 4) outcomes. While humanitarian food assistance is expected to be prioritized to the area and will lead to relatively better outcomes among reached households, disruptions to humanitarian operations are likely due to high residual floodwater and insecurity and access challenges to remote displacement locations.

**From the February 2023 post-harvest period through the typical start of the lean season in May 2023,** food security conditions are expected to deteriorate in both areas due to low or limited availability of food and income sources (including fish and wild foods) and local crop depletion. Additionally, disrupted trade and market operation, rising staple food prices, high residual floodwater, localized insecurity, and cumulative loss in asset base are expected to significantly limit household access to food and income sources. The proportion of households with large to extreme food consumption gaps is expected to increase, with Emergency (IPC Phase 4) outcomes persisting in both areas and some households in Catastrophe (IPC Phase 5). Households in Catastrophe are likely to be those that have experienced multiple displacements, lost all livestock and crop harvest, have limited to no access to markets and humanitarian aid, and live in isolated displacement sites or islands.

## AREAS OF CONCERN

***Fangak, Canal/Pigi, and Akobo of Jonglei: Eastern Plains Sorghum and Cattle LHZ (SS06); and Pibor County in Greater Pibor Administrative Area (GPAA): South-eastern Semi-arid Pastoral livelihood zone (SS05) (Figure 7)***

### *Current Situation*

These four counties in Jonglei and Pibor remain of high concern as households continue to face multiple, often concurrent, shocks, including flooding, conflict, and high food prices, that have undermined the ability of many to access sufficient food. Available household survey data from humanitarian partners and other field assessments such as Inter-agency Rapid Needs Assessments (IRNAs) from July to September, as well as the results from the national IPC conducted in October, confirm the significant deterioration in food security outcomes to Emergency (IPC Phase 4), with some populations in Catastrophe (IPC Phase 5).

### **Fangak and Canal/Pigi**

**Conflict and Insecurity:** Conflict and insecurity remain key drivers of acute food insecurity in these two counties, displacing households and disrupting physical and financial access to food. Fighting along the Upper Nile-northern Jonglei border region escalated from mid-August to mid-October with armed attacks between groups loyal to General Simon's SPLA-IO Kitgwang forces and General Johnson's Agwelek forces displacing 4,506 people from New Fangak to Kurwai of Canal/Pigi and 9,251 people to Old Fangak. In early October, fighting had expanded to the neighboring Malakal and Fashoda of Upper Nile, where

**Figure 7.** Area of concern reference map: Fangak, Canal/Pigi, Akobo of Jonglei; and Pibor Administrative Area



Source: FEWS NET

over 40,000 people have been displaced since the escalation of conflict in mid-August.

**Flooding:** In addition to conflict, flooding has again affected large portions of Fangak and Canal/Pigi, although to a lesser extent than 2021. The above-average rainfall from July to September, coupled with the rising river water levels in the White Nile and Sobat rivers, led to flooding in these low-lying and flood-prone areas of concern that displaced households, destroyed crops, damaged or destroyed households and household assets, disrupted trade routes, and negatively affected market and trade functionality. In Fangak in particular, an [IRNA conducted in August and early September 2022](#) confirmed the devastation caused by floods in Old Fangak, Chotbora, Wanchot, Nonimach, Wangmuk, Nyatuat, Galilee, Pankiir of Old Fangak *payam*, affecting over 17,000 people and destroying around 30,000 houses. Although an official IRNA report is not yet available for Canal/Pigi, approximately 5,000 people are estimated to have been affected by flooding in Atar as of mid-October. While the numbers of flood-affected people in these two counties are lower than in 2021 (estimated at 94,000 in Fangak and 35,000 in Canal/Pigi), the negative impacts of the 2022 floods have compounded the damage done by the 2021 flooding, further decimated crop and livestock production prospects and driving a near collapse of the livelihood system in 2022.

**Crop and Livestock Production:** Field assessments and key informant information align in showing the deleterious effects of years of conflict and flood shocks on crop and livestock production. According to historical data from past WFP-FAO-UNICEF Food Security and Nutrition Monitoring Surveys (FSNMS) and available survey data used during the October 2022 IPC analysis, the proportion of households in these counties engaged in planting has generally declined in recent years. Moreover, the escalation of conflict displaced households and led to field abandonment and disruption of harvesting from August to October in both counties, which contributed to low production in 2022. Although FAO's harvest estimates are not yet available, production in 2022 is likely similar to or lower than last year.

Livestock production systems have been similarly undermined in these areas, with this year's shocks making recovery of animal assets difficult. Key informant information and available survey data, assessed in the context of historical data from past FSNMS surveys, align to suggest that livestock ownership remains very low for the second year in a row in both counties, with particularly steep declines in livestock ownership in Fangak, reflective of the near collapse of the livestock production system. Fangak has experienced its fourth consecutive year of flooding, leading to significant livestock losses and relocation to Pariang of Unity and Ayod and Canal of Jonglei. The decrease in livestock ownership also has negative implications for access to milk, which has declined in parallel in both areas of concern.

**Trade and market system:** With the ability to produce own food significantly disrupted, the importance of market access has increased. However, significant market and trade disruptions have been reported in Fangak and Canal/Pigi due to the most recent escalation of conflict between armed groups in mid-August. The ongoing conflict in the Upper Nile-Jonglei border region has cut off trade routes from Malakal and Bor South, significantly disrupting New Fangak and Atar/Diel markets to the extent that they are operating with minimal to no activity and leaving most households facing access challenges and extremely high food prices. Key informants report displacement of most traders from New Fangak market to Old Fangak, with many traders fearful of trading in these areas due to insecurity and risks to personal safety. Nonetheless, Old Fangak market is recovering from disruptions to trade flows from Malakal and Tonga market of Panyikang, and key informants report a functioning market in Kurwai of Canal/Pigi, albeit with high food prices linked to very limited cereals on the market. The retail price of a *malwa* of sorghum in Old Fangak increased from 4,000 SSP in September to 4,500 SSP in October 2022.

**Fish and wild foods:** Access and availability of fish and wild foods are seasonally at their highest points in these areas, particularly with the above-average rainfall and flooding; however, household access varies depending on the level of standing floodwaters, conflict, and insecurity as well as access to fishing gears. Based on FEWS NET's weekly monitoring and key informants report for mid-October 2022, many households in Fangak are participating in fishing and gathering of wild foods (water lilies), given increased availability.

**Food assistance:** Available evidence shows many households in Fangak and Canal/Pigi are relying on food assistance as a key food source to supplement fish and wild foods, given the loss in crops and livestock production. However, assistance delivery in July-September was lower than planned, reaching only 6 and 5 percent of the county populations in Canal/Pigi and Fangak, respectively. Although key informants have confirmed the distribution of GFD in Old Fangak in October, the assistance reach remains limited in conflict-affected villages in New Fangak and Juaibor, which have not received food assistance since May 2022. Similarly, in Canal/Pigi, key informants report households in Kurwai received GFD between July and August, while in Kuliet, households received it in early October 2022. In Canal/Pigi, food assistance remains lower in Atar 3 due to insecurity. In Fangak, WFP completed distributions in Keew, Paguer, and Old Fangak, while Juaibor, Nyadin, and new Fangak, distributions are suspended due to lack of aircraft.

### **Akobo and Pibor**

While Akobo and Pibor fall into separate livelihood zones – with Akobo mostly within the Eastern Plains Sorghum and Cattle livelihood zone (SS06) with a larger, traditionally agropastoral population and Pibor in the South-eastern Semi-arid Pastoral livelihood zone (SS05) – the similarities in shocks this year and in past years with recurrent flooding and inter-communal violence mean households in these counties remain at risk of acute food insecurity in comparable ways.

**Rainfall performance:** The performance of the June to September rainfall season over Akobo and Pibor varied spatially and temporally. In Akobo, heavy rainfall affected western parts of the county, with the local authority (RRC) reporting 4,840 people affected by floods in 2022 (compared to nearly 10,000 people in 2021). By contrast, prolonged dry spells marked the 2022 season in eastern parts of Akobo and south-eastern pastoral Pibor area. Notably, in the eastern parts of Pibor, households reported that the prolonged dry spells negatively affected crop development and water and pasture availability for livestock at the beginning of the main season in June-July.

**Crop and livestock production:** According to a trend analysis of past FSNMS assessments and available survey data used at the October 2022 IPC analysis in both areas, the proportion of households engaged in planting has declined steadily since 2018. In addition, key informants report high levels of production challenges in both areas. In areas where households planted, key informants have confirmed that the sorghum harvest was completed by early September, with overall crop production in 2022 expected to be similar to last year but significantly below average.

According to available survey data and key informant reports, alongside historical data from past FSNMS surveys, livestock ownership remains low in both counties due to the cumulative negative impacts of repeated flood and conflict events. In Pibor, traditionally a more strictly pastoral zone, the reports of continued low livestock ownership reflects the negative impacts of floods and inter-communal violence and cattle raids in 2020/early 2021 that decimated livestock production. As a result, access to milk has also declined.

**Trade and market system:** In Akobo and Pibor, trade and market functionalities are somewhat better than in Fangak and Canal/Pigi in terms of physical access, though financial access to food is limited by high food prices and the poor economic situation of households. Local market supply has increased with local harvest and supply coming from Ethiopia, but from Juba via air remains very limited. In addition, the trade route linking Akobo West and Waat and on to Lankien and Urur has been cut off due to flooding. Market access is very low in the western Akobo (Buong, Diror and Barmach) linked to persisting floodwaters. Staple food prices are generally very high, though food assistance distributions in August, coupled with some harvests and continuous supply from Ethiopia, facilitated a decline in the retail price of a *malwa* of sorghum from 2,100 SSP in August to 1,330 SSP in late October. Similarly, in Pibor, market access is very low, given the high transportation cost via the Juba and Bor routes to Pibor coupled with the poor feeder road conditions, especially for far remote households in Verteth, Gei, and Kongar. Meanwhile, in Gumuruk, households travel very long distances, often more than one day, to access the market in Pibor. Trade routes linking Pibor from Bor South, Juba, and Kapoeta remain inaccessible. Prices have also been rising, with the retail price of a *malwa* of sorghum increasing from 2,600 SSP in September to 3,500 SSP in October, while dry bean prices per kg rose from 2,500 SSP to 2,700 SSP over the same period.

**Fish and wild foods:** Access to and the availability of fish and wild foods, as in the northern Jonglei areas of concern, are at their seasonally highest points. In Akobo, evidence from key informants and available survey data indicates that access to fish has declined in relatively to past years in both counties; in Pibor, fishing access is low due to water shortages experienced in 2022. In both Akobo and Pibor, access to fish is constrained by distance to fishing grounds and lack of fishing equipment. However, hunting activity is somewhat typically better at this time of the year.

**Food assistance:** Food aid remains pivotal to lessening food consumption gaps, but humanitarians face logistic, security, and funding challenges. Analysis of the GFD and FFA distributions between July and September indicates 18 and 15 percent of the county populations in Akobo and Pibor, respectively, received food aid meeting nearly or above their minimum kilocalorie needs. According to field information and key informant reports, 82 percent in Akobo and 77 percent in Pibor received GFD between June and August. However, since June, most of the areas remained inaccessible since the route linking Pibor and Verteth was cut off by waterweeds, while other villages, including Bubuzen, Gei, and Kongar, were inaccessible due to poor road conditions and insecurity, except for Gumuruk which is accessible via flights.

Available household survey data used for the October 2022 IPC analysis converged with field information to point towards sustained high levels of hunger, marked by poor quantity and dietary quality, in these areas of concern. In addition, household

use of negative coping strategies, including begging, the sale of their last cow or goat, and traveling to other villages in search of food, also remain elevated across the four counties of concern. Broadly, the use of emergency livelihoods coping strategies has remained similar to those reported during the peak of the 2021 lean season. Based on the convergence of all available evidence, Emergency (IPC Phase 4) outcomes exist in all four counties of concern in October, with some populations in Catastrophe (IPC Phase 5).

### *Assumptions*

In addition to the national-level assumptions, the following assumptions apply to these areas of concern:

- Although government's efforts are ongoing to restore peace in Fangak and Canal/Pigi and the neighboring conflict-affected counties of Malakal, Fashoda, and Panyikang, armed clashes between armed groups will likely continue causing further household displacements and disruption to the delivery of assistance, trade flows, and market functioning. However, conflict levels are anticipated to reduce during the projection period, given the additional deployment of unified security forces. In Akobo and Pibor, insecurity and cattle raiding incidents will continue.
- Based on the ICPAC weather forecast, Fangak, Canal/Pigi, and Akobo are likely to experience above-average rainfall during the remainder of the main rainfall season from October to December, likely [to lead to additional flooding](#). Meanwhile, Pibor will receive below-average rainfall. The March-May rains are forecast to be average, though, leading to the recovery of water and pasture availability from May onwards in Pibor.
- Given the overall declines in livestock ownership in all the counties of concern, household access to milk and meat will remain limited, with the greatest seasonal declines between February and May as livestock migrate to distant dry-season grazing areas. Access to milk and meat will likely improve for households who own livestock as livestock start returning from dry season grazing areas to nearer homesteads starting at the end of the projection period in May.
- It is likely that prices will remain above last year and the five-year average through May. In Fangak and Canal/Pigi, food prices will likely remain extremely high given conflict-related disruption to trade flows from Malakal-Fangak and Tonga market, coupled with the lower local harvests and multiple checkpoints along river trade routes to Fangak. Similarly high price trends are likely to persist in Akobo and Pibor due to flood-induced or insecurity-related market access challenges.
- Based on the food assistance distribution plans, WFP plans to reach around 40 percent of the population in Akobo and Pibor with 50-70 percent of their kilocalorie needs through November. Although there is no assistance planned for Fangak and Canal/Pigi during this period, WFP is likely to re-prioritize these counties, given the severity of acute food insecurity based on IPC results. Between December and May, WFP plans to reach 64 percent of the population of Fangak and 40 percent of the population of Pibor with a half ration. In Akobo and Canal/Pigi, 20-30 percent of the population is targeted with a half ration during the same time. The delivery of assistance in all these counties will periodically be disrupted by conflict/insecurity or floodwaters, reducing the level of reach and impact on food security.

### *Most Likely Food Security Outcomes*

Food security will deteriorate in these counties from **October to January** since the majority harvested little to nothing or suffered significant crop and livestock losses. Even amongst households who harvested some food stocks, the harvest is likely to last only up to early January. As such, many will rely on wild foods, fish, and humanitarian food assistance. The worst-off households include the flood-and conflict-affected in Fangak and Canal/Pigi who lost entire livestock or crops to floods or conflict. Emergency (IPC Phase 4) or worse outcomes will persist, with populations in Catastrophe (IPC Phase 5) in Akobo, Fangak, and Pibor. During this period, households will likely continue to use emergency-based coping strategies such as begging from other community members or traveling to other villages or camps in search of foods; however, some households' ability to cope will remain very low or become exhausted, given the inability to continue using some of these strategies.

From **February to May**, which also marks the typical start to the lean season, food security will further deteriorate as most households will have completely exhausted their food stocks and likely depend entirely on fish and wild foods as well as food assistance. Although households will try to rely on financial access to food, extremely high food prices will limit many households from accessing adequate amounts. Many households are likely to face large to extreme consumption gaps, where Emergency (IPC Phase 4) or worse outcomes are likely in all four counties of concern. Significant deterioration in food consumption is expected, especially in Fangak and Canal/Pigi, owing to a near collapse of crop and livestock production. As a



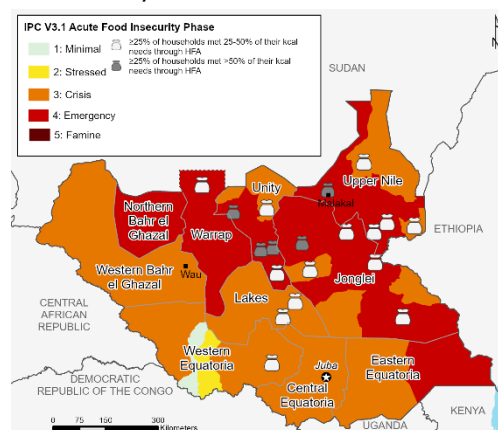
result, some populations in Fangak, Canal/Pigi, and Akobo of Jonglei will be in Catastrophe (IPC Phase 5).

There is a credible, alternative scenario in which Famine (IPC Phase 5) could occur. Fangak and Canal/Pigi are among the areas of most extreme concern for a [risk of Famine \(IPC Phase 5\)](#). Famine (IPC Phase 5) would be likely if conflict and/or weather shocks isolated households from typical food sources and food aid for a prolonged period.

## MOST LIKELY FOOD SECURITY OUTCOMES AND AREAS RECEIVING SIGNIFICANT LEVELS OF HUMANITARIAN ASSISTANCE

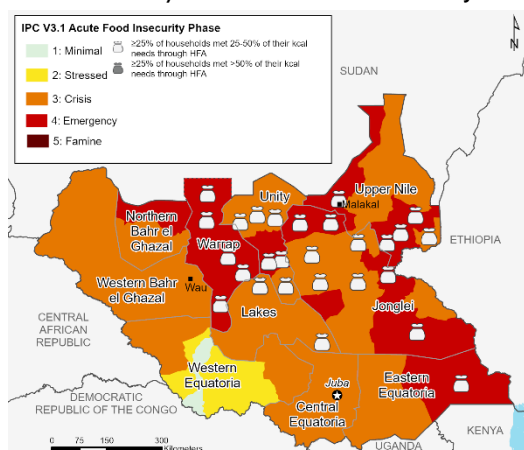
Each of these maps adheres to IPC v3.1 humanitarian assistance mapping protocols and flags where significant levels of humanitarian assistance are being/are expected to be provided. 🍲 indicates that at least 25 percent of households receive on average 25–50 percent of caloric needs from humanitarian food assistance (HFA). 🍲 indicates that at least 25 percent of households receive on average over 50 percent of caloric needs through HFA. This mapping protocol differs from the (!) protocol used in the maps at the top of the report. The use of (!) indicates areas that would likely be at least one phase worse in the absence of current or programmed humanitarian assistance.

Current food security outcomes, October 2022



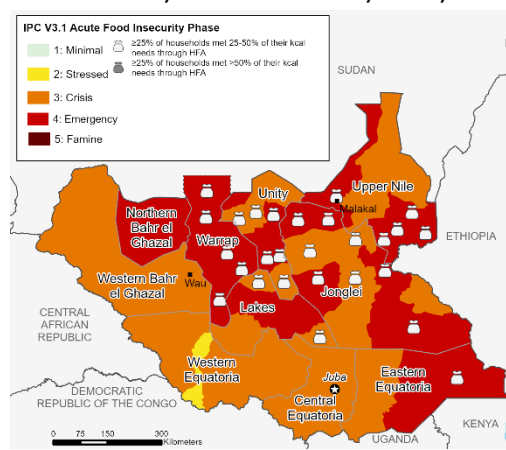
Source: FEWS NET

Projected food security outcomes, October 2022 to January 2023



Source: FEWS NET

Projected food security outcomes, February to May 2023



Source: FEWS NET

FEWS NET's classifications are IPC-compatible. IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

*Recommended citation: FEWS NET. South Sudan Food Security Outlook October 2022 to May 2023: Protracted conflict and floods drive widespread Emergency (IPC Phase 4) during the harvesting period, 2022.*

### ABOUT SCENARIO DEVELOPMENT

To project food security outcomes, FEWS NET develops a set of assumptions about likely events, their effects, and the probable responses of various actors. FEWS NET analyzes these assumptions in the context of current conditions and local livelihoods to arrive at a most likely scenario for the coming eight months. [Learn more here.](#)